

Print Workflow Service DLL Hijacking Vulnerability

How a poorly secured service binary path turned a low-privileged foothold into SYSTEM-level code execution

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Time ... Process Name PID Operation Path Result Detail						
7:26:4...	printworkflowse...	9592	CreateFile	C:\Program Files\Print Workflow Service\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System32\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\helper.dll	NAME NOT FOUND	Desired Access: R...
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7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System32\wbem\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System32\WindowsPowerShell\v1.0\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System32\OpenSSH\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Program Files\OpenSSH-Win64\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Program Files\Git\cmd\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Program Files\Rust stable MSVC 1.95\bin\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Temp\helper.dll	NAME NOT FOUND	Desired Access: R...
7:26:4...	printworkflowse...	9592	CreateFile	C:\Windows\System32\config\systemprofile\AppData\Local\Microsoft\WindowsApps\hel...	NAME NOT FOUND	Desired Access: R...
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7:26:5...	printworkflowse...	9592	CreateFile	C:\Windows\System32\config\systemprofile\AppData\Local\Microsoft\WindowsApps\hel...	NAME NOT FOUND	Desired Access: R...
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Introduction

While building a Windows-focused detection and response training lab, I stumbled across an interesting DLL hijacking issue involving a custom Windows service named Print Workflow Service. The service repeatedly searched for a missing library named helper.dll across multiple writable and semi-trusted paths. What initially looked like noisy Process Monitor telemetry quickly turned into a reliable privilege escalation vector. The deeper I dug into the installer and service configuration, the worse things became.

The Discovery

Watching the service through Process Monitor immediately revealed suspicious behavior. The process continuously attempted to load helper.dll from multiple directories before eventually failing with repeated NAME NOT FOUND results. The service binary operated as NT AUTHORITY\SYSTEM, meaning any malicious DLL successfully loaded by the process would execute with full system privileges.

Installer Misconfigurations

The issue became significantly more severe after reviewing the Print Works installer behavior. During installation, the application assigned overly permissive file permissions to the service directory: C:\Program Files\Print Workflow Service\ Authenticated users retained write access to portions of the directory tree, effectively allowing low-privileged users to modify content loaded by a SYSTEM-level service. Even more concerning, the installer also modified the user PATH environment variable and inserted: C:\Temp into the DLL search path during installation. This dramatically increased the exploitability of the vulnerability since C:\Temp is commonly writable by standard users.

Responsible Disclosure Timeline

The vulnerability was responsibly disclosed to Print Works Inc shortly after discovery. Initial outreach included technical details, proof-of-concept screenshots, and exploitability analysis. Despite acknowledgement of the report, remediation efforts stalled for months. The vendor repeatedly categorized the issue as low severity because it required prior access to the machine. After nearly six months of back-and-forth communication, Print Works Inc finally issued an update addressing the vulnerable DLL loading behavior and tightening installer permissions.

Why This Matters

The frustrating reality is that vendors frequently dismiss local privilege escalation vulnerabilities because they require initial access. That mindset fundamentally misunderstands how modern compromises occur. Attackers rarely start with SYSTEM privileges. Real-world intrusions are chains of weaknesses: Initial Access → Privilege Escalation → Persistence → Lateral Movement DLL hijacking vulnerabilities fit perfectly into that chain.

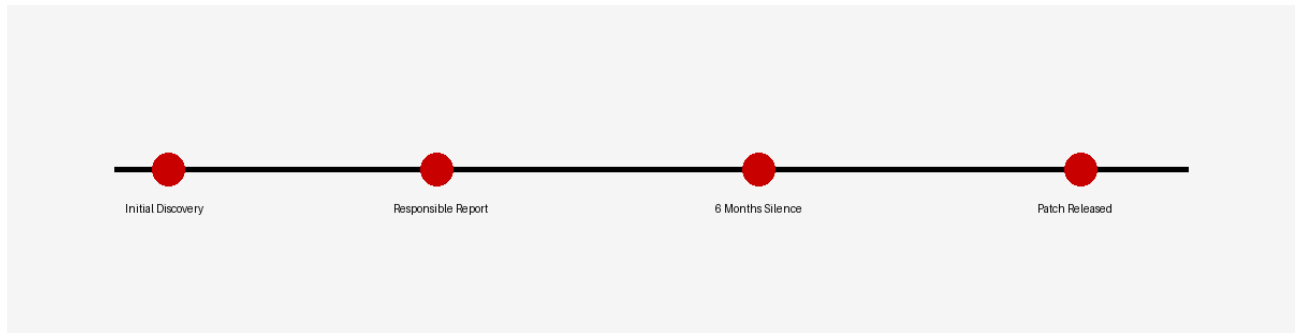
Detection Opportunities

Repeated NAME NOT FOUND telemetry for DLLs remains one of the clearest indicators of potential DLL hijacking opportunities. Useful telemetry includes: - Sysmon Event ID 7 - DLL loads from writable directories - Services probing C:\Temp - Repeated CreateFile failures for DLLs - SYSTEM processes loading unsigned modules

Final Thoughts

This vulnerability serves as an excellent training example because it demonstrates how seemingly minor weaknesses become dangerous once attackers gain a foothold. The combination of: - Unsafe DLL loading - Weak installer permissions - Writable PATH entries - SYSTEM-level service execution created a highly reliable privilege escalation vector that defenders should absolutely monitor for in enterprise environments.

Responsible Disclosure Timeline



Privilege Escalation Flow

